the replication principle

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outline

the idea

get code from others!

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bad excel

- never trust numbers that come from excel
  - no way to find out what happened, there's no code!
    [*]http://andrewgelman.com/2013/04/17/excel-bashing/
- I learned it hard way:
  - my first paper for ecological economics, done in excel
  - reviewers got back after 6mo, i had dozens of excel files
  - couldn't replicate my own results!
- "Talk is cheap. Show me the code"–Linus

replication, replication

- replication=write computer code that will do *everything*
- from raw data (eg FED, IMF) to results (eg regression)
- necessary for science
- otherwise we don't know what happened
- how was it calculated? is there a mistake? who knows?
- pol sci perspective
  [*]http://gking.harvard.edu/files/gking/files/replication.pdf
- IT perspective [*]http://journals.plos.org/plosbiology/article?id=10.1371/journal.pbio.1001745

rules for everyday practice [revisit/stress later!!]

- ◇ once you have coded everything, double check
  - · leave it aside (¿2wks) and check again (does wonders)
  - · show it to other people, post on your website
- ◇ cross-check end output with raw data
  - · eg same values for couple data points? do values make sense?
- ◇ check with alt data? they tell the same story?
  - · I Google/Gemini values/tables/graphs from my output
- ◇ everything has been already studied by others
- ◇ like lit rev, its data rev

outline

the idea

get code from others!

the easiest way to do research in 21st century

- start with code others wrote, and build on their work
- fastest most efficient way to do research
- just email author and ask for code
- even if sas, stata, etc–easy to figure out what's going on and do sth something similar in Py